



Parabolic Dish Solar Collectors for High-Temperature Applications: A Comprehensive Literature Review

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Abstract: Parabolic dish solar collectors (PDSCs) offer exceptional potential for high-temperature solar-energy conversion owing to their point-focus configuration, high concentration capability, and compatibility with advanced thermodynamic and thermal energy storage (TES) technologies. However, their broader deployment remains constrained by coupled optical, thermal, mechanical, materials, and operational challenges that are often addressed independently in the existing literature. This review provides a critical and system-level assessment of recent advances in PDSCs for high-temperature applications, integrating developments in concentrator geometry, optical performance, solar tracking, receiver technologies, thermal management, power conversion, and TES. Particular attention is given to cavity, tubular, volumetric, and phase-change-material-integrated receivers, together with Stirling engines, thermoelectric generators (TEGs), and hybrid conversion architectures. The review further evaluates the role of advanced materials, selective coatings, intelligent tracking, real-time monitoring, numerical modeling, and experimental validation in reducing optical and thermal losses and improving system reliability. Emerging pathways involving hybrid renewable systems, industrial process heat, decentralized energy supply, and solar-driven hydrogen production are critically examined. The analysis identifies material degradation, thermal losses, high capital cost, tracking accuracy, storage limitations, and scalability as major barriers to commercialization. A system-integration perspective is therefore proposed in which optical optimization, receiver design, TES, intelligent control, and power conversion are co-optimized rather than treated as isolated components. Future research should prioritize validated multi-physics modeling, advanced high-temperature materials, intelligent control, scalable thermal storage, techno-economic optimization, and long-duration field demonstration to accelerate the transition of PDSCs toward reliable, dispatchable, and sustainable high-temperature energy systems.

Keywords: Parabolic dish solar collectors; Concentrated solar power; High-temperature applications; Thermal energy storage; Stirling engines; Hybrid solar systems

1 Introduction

Parabolic dish solar collectors (PDSCs) can achieve high solar concentration ratios and operating temperatures, making them attractive for electricity generation and high-temperature industrial processes. Their performance, however, depends strongly on the accuracy of the parabolic geometry, reflector surface quality, and solar tracking. Small deviations in curvature, surface slope, or alignment can broaden the focal spot and increase optical losses. Consequently, precision manufacturing, high-reflectivity coatings, accurate tracking, and ray-tracing-based optical optimization are essential for maximizing the solar flux delivered to the receiver [1, 2]. The overall performance of a PDSC is determined by the integrated operation of the reflector, receiver, tracking system, power-conversion unit, and thermal energy storage (TES) system. The reflector concentrates solar radiation, the receiver converts it into useful thermal energy, while the power-conversion system determines the efficiency of electricity generation. TES can further improve system flexibility by decoupling solar collection from energy utilization. In this context, Stirling engines and phase change materials (PCMs) represent promising technologies for hybrid PDSC systems. Al-Hashmi [3] experimentally investigated a parabolic dish integrated with a Stirling engine and PCM storage, demonstrating the potential of combining concentrated solar power (CSP) generation with thermal storage. The effectiveness of PCM

storage depends on key thermophysical properties, including melting temperature, latent heat, thermal conductivity, thermal stability, and cycling durability [4]. Proper matching of these properties with the collector operating temperature is therefore essential as shown in Figure 1. More recently, Al-Hashmi et al. [5] demonstrated the benefits of integrating a PDSC, Stirling engine, and PCM-based TES to improve energy availability and operational flexibility. Overall, integrating PDSCs with efficient power conversion and TES provides a promising pathway for reducing solar intermittency and extending energy availability beyond periods of direct solar irradiation, particularly in regions with high direct normal irradiance (DNI), such as Oman [5].

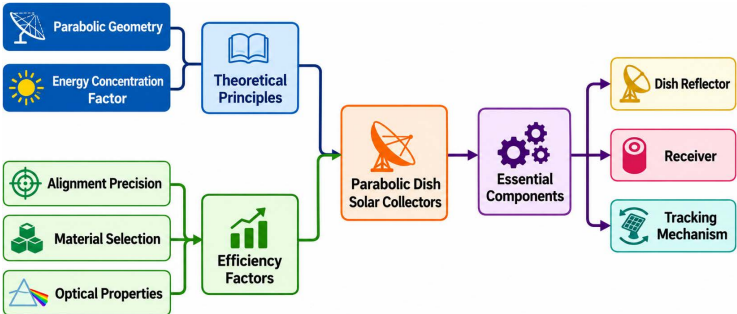


Figure 1. Parabolic dish solar collectors (PDSCs): design, components, and efficiency

2 Design Principles and Components

The dish reflector is the most visually prominent component of the system, responsible for collecting and concentrating sunlight. It is typically constructed from a lightweight framework covered with reflective materials, such as polished aluminum or silver-coated glass. The precision with which the parabolic surface is fabricated directly impacts the system’s ability to focus sunlight accurately onto the receiver. The studies reported in [6, 7] emphasize that achieving an ideal parabolic shape with minimal surface imperfections is critical for maximizing the optical efficiency of the collector. The choice of reflective material is equally significant, as it affects the system’s durability and long-term performance. Materials with high reflectivity and resistance to environmental degradation, such as oxidation or corrosion, are preferred. The study reported in [8] highlights the importance of material advancements in enhancing the lifespan of parabolic dish collectors (PDCs) as shown in Figure 2.

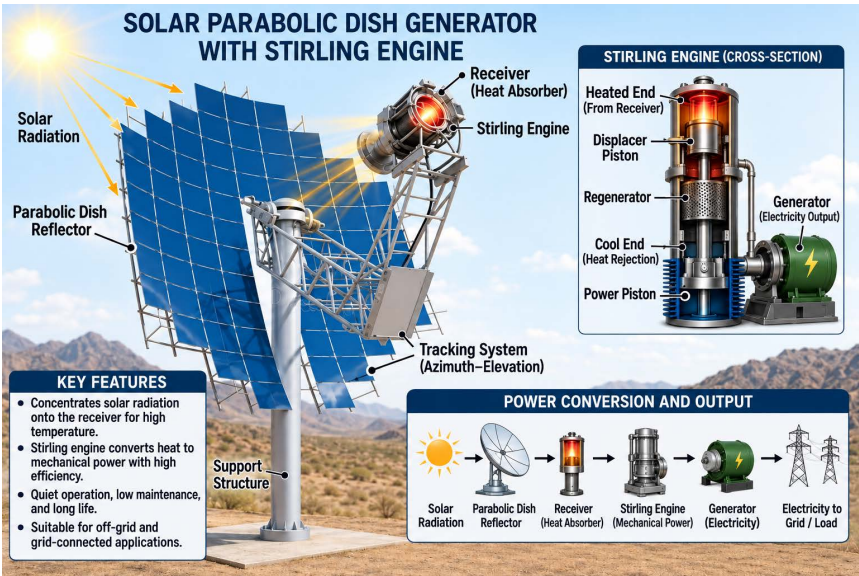


Figure 2. Parabolic dish solar collectors (PDSCs) with stirling engine

2.1 System Architecture of Parabolic Dish Solar Collectors

For instance, the development of advanced coatings that can withstand high temperatures and harsh environmental conditions has contributed to improving the reliability of these systems. The receiver is the component where concentrated solar energy is absorbed and converted into thermal energy. Depending on the application, different receiver designs, such as cavity receivers and tubular receivers, may be employed. Notes that the thermal performance

of the receiver is influenced by factors such as its geometry, material composition, and heat transfer properties. Receivers made from high-conductivity materials, such as copper or stainless steel, are commonly used to ensure efficient heat transfer from the concentrated sunlight to the working fluid. The design of the receiver also considers thermal losses due to convection and radiation. To minimize these losses, receivers are often equipped with insulating materials or enclosed within a vacuum-sealed chamber. Discuss the integration of PCMs within receivers to enhance their thermal storage capacity [8–11]. By storing excess thermal energy during periods of high solar irradiance, PCMs enable the system to maintain a consistent energy output even under fluctuating sunlight conditions as shown in Figure 2 and Table 1.

Table 1. System architecture and design principles of parabolic dish solar collectors (PDSCs)

Component	Design Principle	Function	Key Considerations
Parabolic reflector (dish)	Parabolic geometry for point-focus solar concentration	Concentrates Direct Normal Irradiance (DNI) onto receiver	High reflectivity, surface accuracy slope error, durability
Receiver (absorber)	Located at focal point to maximize energy absorption	Converts concentrated radiation into thermal energy	Cavity/tubular design, selective coating, thermal losses
Solar tracking system	Dual axis tracking for continuous sun alignment	Maintains optimal incident angle	Tracking accuracy, control algorithms, reliability
Support structure	Structural rigidity with minimal deformation	Maintains dish geometry and stability	Wind loads, thermal expansion, material strength
Power conversion unit	High-temperature thermodynamic efficiency	Converts thermal energy to electricity	Stirling engine coupling, efficiency, cooling
Thermal energy storage (optional)	Sensible/latent heat storage integration	Enables dispatchability and load shifting	PCM selection, thermal conductivity, cycling stability
Control & instrumentation	Real-time monitoring and feedback control	Optimizes performance and system safety	Sensors, automation, fault detection

Solar tracking is essential for maintaining the alignment between a PDSC and the sun, thereby ensuring that concentrated radiation remains focused on the receiver. Even small tracking errors can shift the focal spot and significantly reduce the intercepted solar flux, particularly at high concentration ratios. PDSCs commonly employ single-axis or dual-axis tracking systems. While single-axis systems are simpler and less expensive, dual-axis trackers provide more accurate alignment by independently controlling azimuth and elevation angles, making them more suitable for high-temperature applications [12, 13]. Tracking accuracy depends on sensor precision, actuator response, structural rigidity, and control algorithms. Mechanical backlash, wind loading, thermal deformation, and sensor errors can introduce additional angular deviations. Consequently, modern systems increasingly employ closed-loop control, optical sensors, and real-time feedback to minimize tracking errors and maintain stable solar concentration [14]. Artificial intelligence and machine-learning approaches offer further opportunities for predictive tracking and adaptive correction under variable environmental conditions [15]. Material selection also strongly influences PDSC efficiency and durability. Reflector materials should provide high solar reflectivity, low surface roughness, mechanical stability, and resistance to ultraviolet radiation, humidity, dust, and temperature fluctuations. Advanced reflective coatings can reduce optical degradation and extend service life [16, 17]. Similarly, receiver materials must withstand high temperatures, thermal cycling, oxidation, and concentrated heat fluxes. High-temperature alloys, ceramics, and selective coatings can enhance thermal stability while increasing solar absorptance and reducing radiative losses [18, 19]. Optimizing tracking accuracy, reflector durability, and receiver materials simultaneously is essential for achieving reliable high-temperature PDSC operation. Future systems should integrate predictive tracking, real-time monitoring, advanced coatings, and high-temperature materials to improve efficiency, reliability, and long-term performance.

2.2 Geometry and Optical Performance of Parabolic Dish Solar Collectors

The geometrical configuration of a PDSC is fundamental to its optical efficiency, concentration capability, and high-temperature performance. The principal parametersrim angle, aperture diameter, focal length, and focal-to-aperture ratio determine the concentration ratio, focal position, receiver size, and distribution of concentrated solar flux ash sow in Figure 3. These parameters must be optimized collectively because increasing one parameter may improve solar collection while simultaneously increasing structural, tracking, or optical requirements. The rim angle affects concentration capability and flux distribution, whereas a larger aperture increases the intercepted solar power

but requires greater structural rigidity and tracking accuracy [12, 14, 20]. Similarly, focal length influences the concentration level and receiver configuration; shorter focal lengths can enhance concentration but may increase sensitivity to slope and tracking errors [21, 22]. In practical systems, optical performance is reduced by reflector surface roughness, manufacturing tolerances, slope errors, tracking inaccuracies, shading, blocking, and atmospheric attenuation. These factors broaden the focal spot and reduce the fraction of concentrated radiation intercepted by the receiver. High-quality reflective materials, precision fabrication, and accurate two-axis tracking are therefore essential, particularly for large-aperture systems exposed to wind loading and thermal deformation [23, 24]. Optical efficiency, intercept factor, concentration ratio, and receiver flux distribution provide complementary indicators of system performance. Advanced ray-tracing and Monte Carlo simulations enable these losses to be quantified and support optimization of reflector and receiver configurations [21, 25, 26]. Recent developments have shifted PDSC design toward segmented, non-uniform, and adaptive reflector geometries. Segmentation can simplify manufacturing and maintenance while allowing individual alignment correction; non-uniform configurations can improve flux distribution and receiver utilization; and adaptive reflectors can dynamically compensate for changing solar and operating conditions [27, 28]. These approaches demonstrate the transition from conventional geometric optimization toward flexible and intelligent concentrator architectures as shown in Figure 3.

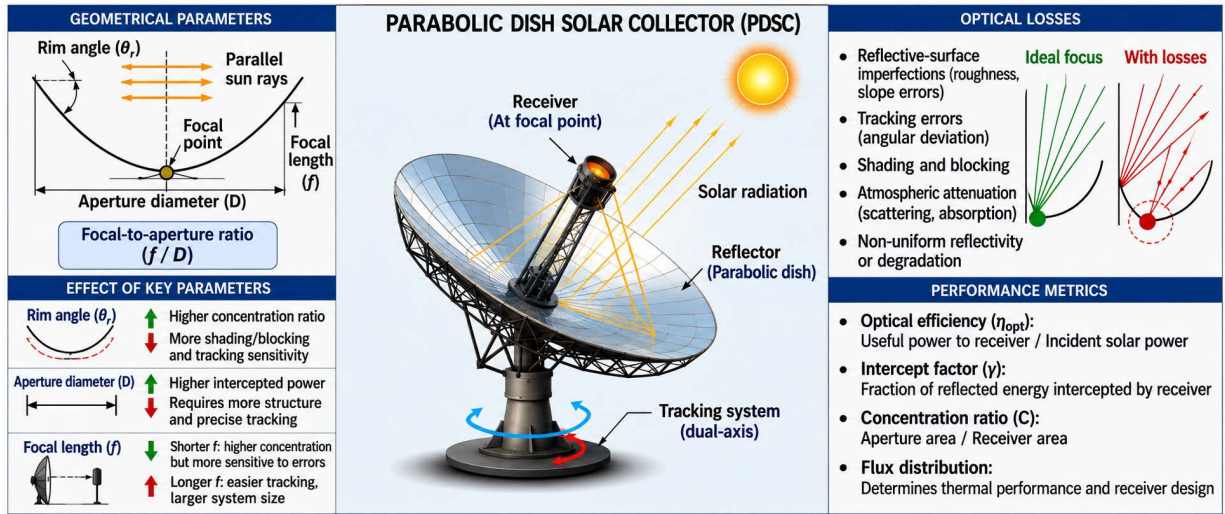


Figure 3. Geometrical parameters, optical losses, and performance optimization of parabolic dish solar collectors (PDSCs)

The effective PDSC design requires integrated optimization of geometry, optical quality, tracking, receiver characteristics, and structural stability. Advanced reflective coatings, precision optical metrology, adaptive tracking, and machine-learning-based control offer additional opportunities to minimize optical losses [29]. Future research should therefore combine high-fidelity optical modeling with experimental validation and coupled optical thermal structural analysis. Such an integrated strategy is particularly important for high-temperature applications, where improved optical concentration directly enhances receiver temperature, thermal efficiency, and the performance of downstream power-conversion and TES systems.

2.3 Receiver Configurations for Enhanced Thermal Efficiency

The receiver in a PDSC plays a critical role in capturing and converting concentrated solar energy into thermal energy. Various receiver designs have been developed to maximize thermal efficiency and cater to specific high-temperature applications. Two prominent configurations are cavity receivers and tubular receivers, each with unique attributes that influence their performance.

2.3.1 Cavity receivers

Cavity receivers are enclosed systems designed to minimize thermal losses by reducing radiation and convection. Their design typically includes a hemispherical or cylindrical cavity with an aperture to allow concentrated solar radiation to enter. Once the solar radiation is absorbed, heat is retained within the cavity, ensuring higher thermal efficiency. The geometry of the cavity is crucial; a smaller aperture relative to the cavity size reduces heat loss but may limit the amount of solar energy entering the receiver. Emphasized that cavity receivers are particularly effective in applications requiring prolonged thermal retention, such as industrial process heating [30, 31]. Additionally, cavity receivers excel in applications where thermal energy needs to be stored or distributed consistently over time as show in Figure 4 and Table 2.

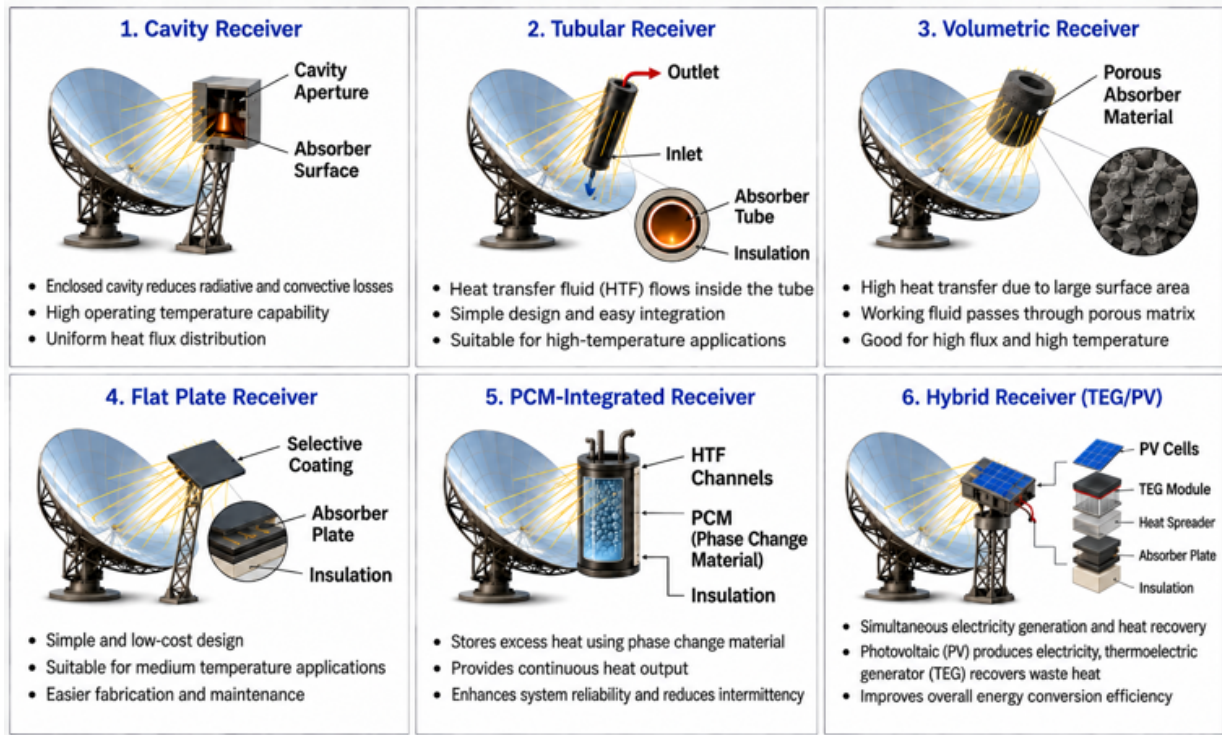


Figure 4. Types of receivers for parabolic dish solar collectors (PDSCs)

Table 2. Parabolic dish solar collectors (PDSCs) with different thermal receiver configurations [32]

Receiver Type	Design Description	Operating Temperature Range (°C)	Thermal Efficiency	Advantages	Limitations	Typical Applications
Cavity receiver	Enclosed absorber with small aperture to reduce radiative losses	500–1000	High	Low heat losses, high absorption	Complex design, cooling challenges	Stirling engines, high-temperature power generation
Tubular receiver	Array of tubes carrying HTF	300–700	Moderate–high	Simple design, easy HTF circulation	Higher convective/radiative losses	Process heat, hybrid CSP systems
Volumetric receiver	Porous structure allowing radiation penetration and volumetric absorption	700–1200	Very high	Uniform temperature, reduced surface losses	Material durability, high cost	Advanced CSP, solar fuels
Flat Plate receiver	Planar absorber at focal region	200–400	Moderate	Low cost, simple fabrication	Limited temperature capability	Low-temperature thermal applications
PCM-integrated receiver	Receiver combined with phase change material for storage	300–800	High (effective)	Thermal storage, load shifting	PCM degradation, thermal conductivity limits	Dispatchable power, hybrid systems
Hybrid receiver (TEG/PV)	Combines thermal receiver with TEGs or PV cells	200–600	Moderate–high	Multi-energy output	System complexity, thermal management	Hybrid power generation

Note: HTF = heat transfer fluid; CSP = concentrated solar power; TEGs = thermoelectric generators; PV = photovoltaic.

2.3.2 Tubular receivers

Tubular receivers as shown in Figure 3, on the other hand, consist of a series of tubes through which a heat transfer fluid (HTF) circulates. These receivers directly absorb solar radiation onto the tube surfaces, transferring heat to the HTF. Noted that tubular receivers are particularly suited for systems requiring rapid heat transfer and high fluid flow rates. They are often coated with selective materials to enhance absorption while minimizing re-radiation losses. The efficiency of tubular receivers largely depends on their configuration, material properties, and the type of HTF used. Commonly used HTFs include molten salts, synthetic oils, and water-steam mixtures, each tailored to specific temperature ranges and operational needs. The thermal efficiency of receivers is often evaluated using metrics such as absorption efficiency, heat transfer rate, and thermal losses [30, 33, 34]. highlighted that the choice of receiver design significantly impacts these metrics. Cavity receivers typically achieve higher absorption efficiency due to their enclosed design, which minimizes radiation losses. However, they may experience challenges related to convection losses if not adequately insulated. Tubular receivers, while efficient in rapid heat transfer, may suffer from higher radiation losses due to their exposed surfaces. Experimental studies have demonstrated that cavity receivers can achieve absorption efficiencies exceeding 90% under optimal conditions, while tubular receivers often range between 70% and 85% depending on their material and coating properties. The choice of receiver design should align with the specific thermal requirements of the application, considering factors like operating temperature, HTF compatibility, and system integration. The materials used in receiver construction are pivotal in determining their thermal conductivity, durability, and overall performance. High temperature applications demand materials that can withstand extreme thermal stresses while maintaining efficient heat transfer properties [29, 35].

2.4 Metallic Materials

Metallic materials such as stainless steel and copper are commonly used due to their high thermal conductivity and structural integrity. Stainless steel is particularly favored for its corrosion resistance and ability to withstand temperatures above 500 °C, making it suitable for tubular receivers. Copper, while offering excellent thermal conductivity, is often used in smaller-scale applications due to its higher cost and susceptibility to oxidation at elevated temperatures [36, 37].

2.4.1 Ceramic materials

Ceramic materials, including silicon carbide and alumina, have gained popularity for their high-temperature resilience and low thermal expansion. Noted that ceramic receivers are ideal for applications requiring temperatures above 800 °C, such as hydrogen production and advanced industrial processes. Despite their advantages, ceramics may face challenges related to brittleness and manufacturing complexity [36, 38, 39].

2.4.2 Active cooling systems

Active cooling involves the use of external systems to regulate the receiver's temperature. For instance, forced convection systems use pumps or fans to circulate air or liquid across the receiver surface, dissipating excess heat. Highlighted the effectiveness of active cooling in maintaining stable operating temperatures in tubular receivers. While active cooling systems are highly effective, they require additional energy inputs and may increase system complexity [40].

2.4.3 Passive cooling systems

Passive cooling relies on natural heat dissipation mechanisms, such as radiation and convection, to regulate receiver temperatures. Noted that passive cooling systems are particularly advantageous in remote or off-grid applications where energy resources are limited. Designs incorporating fins or heat sinks can enhance passive cooling efficiency by increasing the surface area for heat dissipation [41].

2.4.4 Phase change materials

The integration of PCMs into receiver designs represents a significant innovation in thermal management. PCMs absorb and release heat during phase transitions, providing a stable temperature regulation mechanism. Demonstrated that PCM-based cooling systems could reduce peak temperatures in cavity receivers by up to 30%, extending their operational lifespan. However, the selection of PCMs must consider factors like melting point, thermal conductivity, and compatibility with receiver materials. Several experimental studies have showcased advancements in receiver designs and cooling mechanisms as shown in Table 3. For instance, Conducted a comparative analysis of cavity and tubular receivers in a high-temperature solar collector system. The study revealed that cavity receivers achieved a 15% higher thermal efficiency compared to tubular receivers under similar operating conditions, primarily due to reduced radiation losses. Additionally, the integration of selective coatings in tubular receivers improved their absorption efficiency by 10%, highlighting the importance of material optimization. Another study, explored the use of PCMs in tubular receivers for industrial process heat applications. The findings indicated that PCM-based cooling systems enhanced temperature stability, reducing thermal stresses and improving overall system reliability. These results underscore the potential of innovative cooling mechanisms in advancing receiver performance [42, 43].

Table 3. Parabolic dish solar collectors (PDSCs) with phase change material (PCM) energy storage

PCM Type	Typical PCM Materials	Melting Temperature Range (°C)	Storage Mechanism	Key Advantages	Key Challenges	Typical Applications
Organic PCMs	Paraffin wax, stearic acid	50–300	Latent heat	Chemical stability, no supercooling, non-corrosive	Low thermal conductivity, flammability	Small-scale PDSCs, low-medium temperature storage
Inorganic PCMs	Nitrates, chlorides, carbonates	300–700	Latent heat	High thermal conductivity, high storage density	Corrosion, phase segregation	High-temperature PDSC-Stirling systems
Metallic PCMs	Al–Si, Mg–Zn alloys	500–900	Latent heat	Very high thermal conductivity, compact storage	High cost, oxidation	Advanced high-temperature PDSCs
Composite PCMs	PCM + graphite/metal foam	100–700	Latent + enhanced conduction	Improved heat transfer, stable cycling	Increased complexity and cost	Fast charging/discharging systems
Encapsulated PCMs	Micro/macroencapsulated PCMs	100–800	Latent heat	Leakage prevention, modular design	Encapsulation degradation	Modular thermal storage units
PCM-integrated receivers	PCM embedded in cavity/tubular receiver	300–800	Direct thermal buffering	Reduced temperature fluctuations	Structural and thermal stress	Dispatchable PDSC power generation

3 Performance Optimization Strategies

3.1 Heat Transfer Mechanisms and Thermal Management

The efficiency of PDSCs hinges significantly on the effective transfer and management of heat. Heat transfer in these systems is influenced by three primary mechanisms: conduction, convection, and radiation. Each mechanism plays a critical role in ensuring the system's thermal energy is utilized effectively for high-temperature applications. Additionally, innovative cooling strategies both active and passive are essential to maintaining optimal thermal efficiency and preventing system degradation. Material advancements and system modifications have further refined heat transfer processes, leading to improved performance metrics in experimental studies. This section delves into the theoretical foundations of heat transfer within PDSCs, explores advanced cooling techniques, and discusses experimental data highlighting advancements in thermal management. Heat transfer within PDSCs involves the interplay of conduction, convection, and radiation, each contributing uniquely to the system's thermal performance. Conduction refers to the transfer of heat through solid materials, such as the receiver and support structures, which are often exposed to high temperatures. The selection of materials with high thermal conductivity is crucial to minimizing thermal resistance and ensuring efficient heat flow. For instance, metals like copper and aluminum are frequently used due to their superior conductive properties [44–46].

Convection, on the other hand, occurs when heat is transferred through a fluid medium, such as air or a HTF. Forced convection, achieved through the use of pumps or fans, enhances the heat transfer rate by increasing the fluid velocity over heated surfaces as shown in Figure 5. Natural convection, driven by temperature-induced density differences, also contributes to heat dissipation but is often less efficient than its forced counterpart. Emphasize the importance of optimizing the flow characteristics of HTFs to achieve superior thermal performance in high-temperature applications. Radiation, the third mechanism, involves the transfer of energy through electromagnetic waves. In PDSCs, radiation plays a pivotal role in the initial energy capture, as solar radiation is concentrated onto the receiver. However, thermal radiation losses from the receiver can significantly impact overall efficiency. Highlight the importance of minimizing radiative losses through the use of selective coatings and advanced receiver designs that reduce emissivity while maximizing absorptivity. Cooling strategies are integral to maintaining the thermal efficiency of PDSCs, particularly in high-temperature applications where excessive heat can lead to material degradation and system inefficiencies. Cooling mechanisms can be broadly categorized into active and passive

methods [11, 47–49].

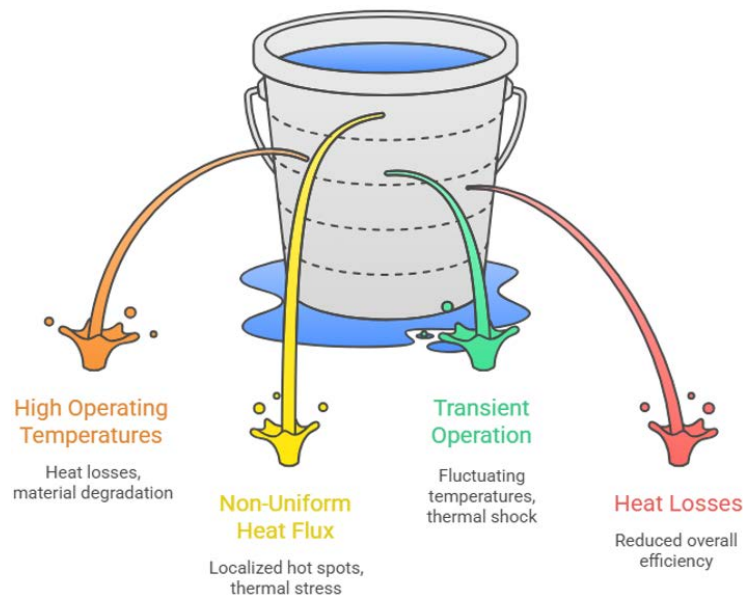


Figure 5. Parabolic dish solar collectors (PDSCs) performance optimization strategies

3.2 Active Cooling Strategies

Active cooling involves the use of external power sources to drive cooling processes. Common techniques include forced air cooling and liquid cooling, which are particularly effective in high-temperature scenarios. Forced air cooling employs fans to increase airflow over heated surfaces, thereby enhancing convective heat transfer. Liquid cooling, on the other hand, uses fluids such as water or specialized HTFs to absorb and transport heat away from the receiver. PCMs absorb heat during phase transitions, providing a buffer against temperature spikes and ensuring stable thermal performance. Another innovative active cooling approach involves the use of thermoelectric coolers (TECs). These devices exploit the Peltier effect to create a temperature gradient, enabling heat to be actively pumped away from the receiver. While TECs offer precise temperature control, their energy consumption and cost can be limiting factors [50].

3.3 Passive Cooling (Thermal Management and Heat-Transfer Enhancement)

Passive cooling strategies are increasingly attractive for PDSCs because they dissipate excess thermal energy with minimal parasitic power consumption. These approaches exploit radiative heat transfer, natural convection, optimized receiver geometry, and thermally functional materials to regulate receiver temperature. Effective thermal management is particularly important in high-temperature PDSCs, where excessive temperatures can accelerate material degradation and increase radiative and convective losses. Therefore, passive cooling should be considered as an integral component of receiver and system-level design rather than an isolated heat-rejection strategy.

Radiative cooling is an important passive mechanism in high-temperature receivers as shown in Figure 6. It relies on thermal radiation from the receiver toward the surrounding environment, with its effectiveness governed by surface emissivity, temperature, and radiative view factors. Advanced coatings can modify these properties and improve the balance between solar absorption and infrared emission. Highlighted the potential of advanced surface treatments for improving thermal management. However, the optimum radiative characteristics depend strongly on operating temperature and receiver configuration. Consequently, coating selection should be integrated with optical and thermal optimization rather than based solely on nominal absorptivity and emissivity.

Receiver geometry also plays a critical role in passive heat dissipation. Configurations that promote buoyancy-driven airflow and increase effective heat-transfer area can enhance natural convection without additional electrical consumption. Cavity receivers are particularly attractive because they can reduce direct exposure of hot surfaces while maintaining effective internal heat transfer. Emphasized the importance of incorporating passive cooling features into receiver design. Nevertheless, excessive surface exposure may increase convective losses, indicating a fundamental trade-off between heat dissipation and preservation of useful thermal energy.

Experimental research has also investigated advanced HTFs, coatings, and receiver configurations. Nanofluids containing metal-oxide nanoparticles, such as aluminum oxide and copper oxide, can improve effective thermal conductivity and convective heat transfer compared with conventional HTFs. However, their practical application

remains constrained by nanoparticle agglomeration, increased viscosity, pressure-drop penalties, erosion, long-term stability, and cost. Similarly, selective receiver coatings can increase solar absorptance while reducing infrared emissivity, thereby lowering radiative losses [51]. Their long-term performance, however, depends on resistance to oxidation, thermal cycling, contamination, and high-temperature degradation. Receiver geometry provides an additional pathway for improving thermal performance. Demonstrated that optimized cavity configurations can reduce convective and radiative losses by limiting direct exposure of the absorber. However, reported efficiency improvements should be compared cautiously because differences in concentration ratio, operating temperature, ambient conditions, and experimental methodology can significantly influence measured performance.

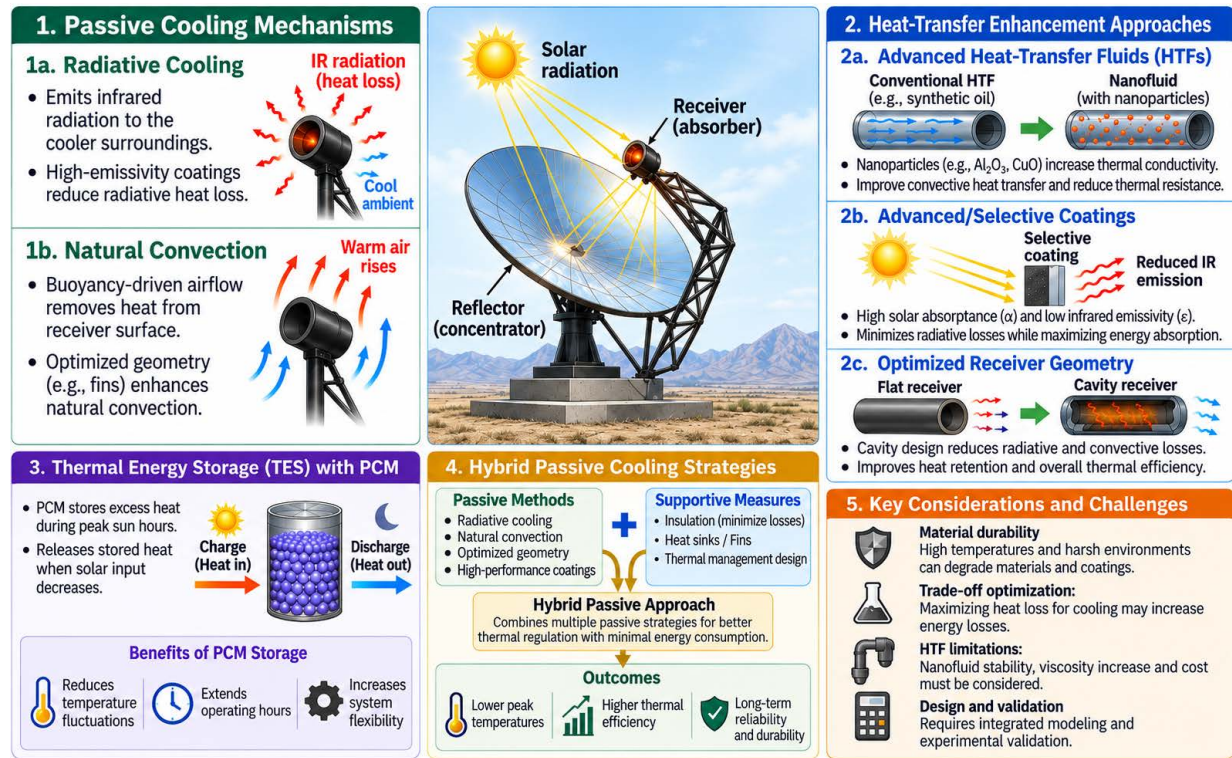


Figure 6. Passive cooling and heat-transfer enhancement strategies for parabolic dish solar collectors (PDSCs)

TES provides complementary thermal management by storing excess heat during periods of high solar irradiance and releasing it when solar input decreases. PCMs are particularly promising because their latent heat enables high energy storage density with relatively stable operating temperatures. In PDSCs, PCM integration can reduce temperature fluctuations and extend useful energy delivery. Nevertheless, limited thermal conductivity, phase segregation, supercooling, material degradation, and containment requirements remain important barriers. Challenge is balancing cooling effectiveness with system efficiency. Active cooling provides strong temperature control but introduces parasitic energy consumption, whereas passive cooling may be insufficient under extreme operating conditions. Hybrid cooling systems combining passive and selectively activated active cooling therefore offer a promising compromise. In parallel, corrosion-resistant materials and durable coatings are essential for long-term reliability under high-temperature and harsh environmental conditions [50–52].

Advanced computational fluid dynamics (CFD), radiation, and conjugate heat-transfer models can further optimize receiver geometry, temperature distribution, and thermal losses. The value of computational modeling for improving solar receiver performance. Future research should therefore integrate validated multi-physics simulations with experimental testing, durable high-temperature materials, intelligent cooling control, and PCM-based TES. Such integrated thermo-optical and materials optimization is essential for improving PDSC efficiency, reliability, durability, and commercial viability in high-temperature applications.

4 Power Conversion Systems for High-Temperature Applications

The integration of advanced power conversion systems with PDSCs has been instrumental in harnessing solar energy for high-temperature applications. These systems, which include Stirling engines, thermoelectric generators (TEGs), and other innovative technologies, are essential for converting concentrated solar energy into usable power. In this discussion, we explore the theoretical underpinnings of these technologies, examine their integration challenges, and highlight recent innovations that have enhanced their efficiency and reliability. Power conversion technologies

play a critical role in the functionality of PDSCs. Among the most promising technologies for high temperature solar applications are Stirling engines and TEGs. These systems are uniquely suited to withstand and utilize the high temperatures generated by parabolic dish concentrators.

4.1 Stirling Engines

Stirling engines are external combustion engines that operate on the principle of cyclic compression and expansion of a gas at different temperature levels, resulting in mechanical work. These engines are particularly well-suited for integration with PDSCs due to their ability to convert concentrated solar heat into mechanical energy efficiently. The high thermal efficiency of Stirling engines, which can exceed 40%, makes them a preferred choice for solar power generation. Moreover, their compatibility with various heat sources, including solar energy, enhances their adaptability in different operational contexts. Recent advancements in Stirling engine design have focused on improving their thermal efficiency and operational stability as shown in Figure 7. For instance, researchers have developed regenerative Stirling engines with enhanced heat exchangers, which maximize heat transfer and minimize thermal losses. These innovations have significantly improved the performance of Stirling engines in high-temperature environments, making them an integral component of solar power systems [53, 54].

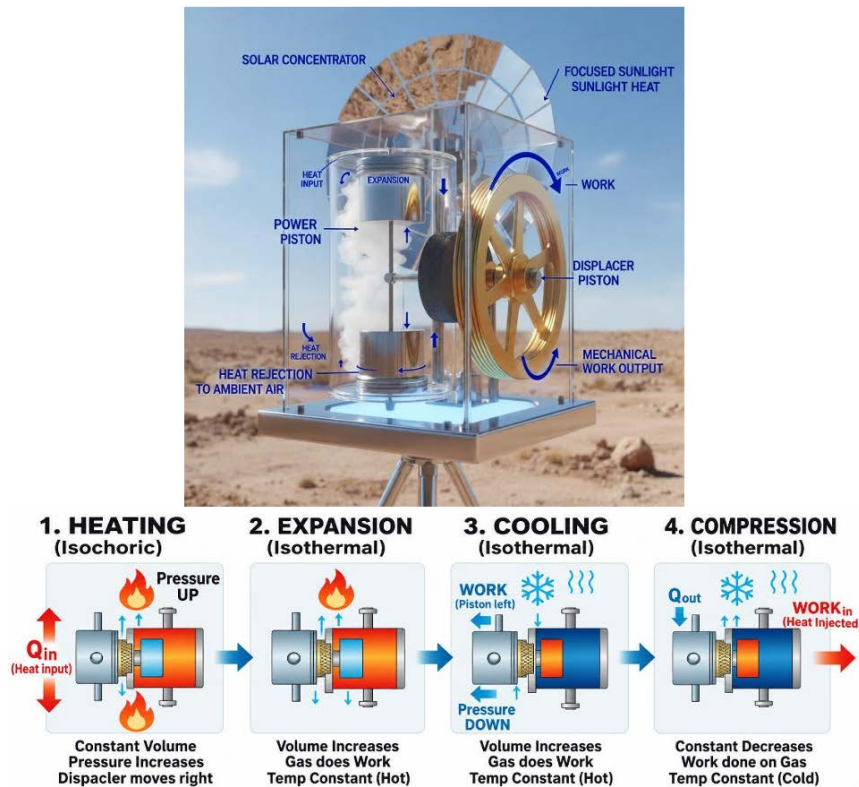


Figure 7. Components and working principle of a solar Stirling engine and the four processes of the Stirling cycle

4.2 Thermoelectric Generators

TEGs are another promising technology for power conversion in parabolic dish systems. These devices convert heat directly into electricity using the Seebeck effect, which occurs when a temperature gradient is applied across a thermoelectric material. TEGs are particularly advantageous for solar applications due to their solid-state nature, which eliminates the need for moving parts and reduces maintenance requirement [55, 56]. Additionally, TEGs can operate over a wide range of temperatures, making them suitable for high-temperature solar applications. However, the efficiency of TEGs has traditionally been limited by the thermoelectric materials used. Recent research has focused on developing advanced materials with higher thermoelectric performance, such as nanostructured and composite materials. These materials exhibit higher Seebeck coefficients and lower thermal conductivity, which enhance the overall efficiency of TEGs. For example, Demonstrated that incorporating nanostructured thermoelectric materials into TEGs can improve their efficiency by up to 15%, making them more competitive with other power conversion technologies. While power conversion systems like Stirling engines and TEGs offer significant advantages, their integration with PDSCs presents several challenges. These challenges include thermal losses, operational stability, and system compatibility.

4.3 Thermal Losses

One of the primary challenges in integrating power conversion systems with PDCs is minimizing thermal losses. Parabolic dish systems concentrate solar energy onto a receiver, where it is converted into heat. However, a significant portion of this heat can be lost through conduction, convection, and radiation, reducing the overall efficiency of the system. To address this issue, researchers have developed advanced receiver designs with improved thermal insulation and selective coatings that minimize heat losses while maximizing energy absorption.

4.4 Operational Stability

Another challenge is ensuring the operational stability of power conversion systems under varying solar conditions. Parabolic dish systems are highly dependent on solar irradiance, which can fluctuate throughout the day and across seasons. These fluctuations can impact the performance of power conversion systems, leading to inefficiencies and potential system failures. To mitigate these issues, researchers have developed advanced control algorithms and real-time monitoring systems that optimize the operation of power conversion technologies under dynamic conditions.

4.5 System Compatibility

The integration of power conversion systems with PDCs also requires careful consideration of system compatibility. For example, Stirling engines require precise alignment with the focal point of the parabolic dish to ensure optimal heat transfer. Similarly, TEGs must be designed to withstand the high temperatures and thermal stresses associated with concentrated solar energy. Researchers have addressed these challenges by developing modular and scalable power conversion systems that can be easily integrated with different types of PDCs. Recent advancements in power conversion systems have focused on improving their efficiency, reliability, and adaptability for high-temperature solar applications. These innovations include the development of hybrid systems, advanced materials, and novel design approaches.

4.6 Hybrid Systems

Hybrid power conversion systems combine multiple technologies to optimize energy conversion efficiency. For example, some systems integrate Stirling engines with TEGs, leveraging the strengths of both technologies. This approach allows for more efficient utilization of the heat generated by PDCs, as TEGs can convert residual heat into electricity after the Stirling engine has extracted the majority of the energy. Hybrid systems have demonstrated significant performance improvements in experimental studies, with some configurations achieving overall efficiencies of up to 50%.

4.7 Advanced Materials

The development of advanced materials has also played a crucial role in enhancing the performance of power conversion systems. For instance, researchers have developed high-temperature-resistant materials for Stirling engine components, which improve their durability and thermal efficiency. Similarly, the use of nanostructured thermoelectric materials has significantly enhanced the performance of TEGs, as previously mentioned. These materials are not only more efficient but also more cost-effective, making them a viable option for large-scale solar power installations.

4.8 Novel Design Approaches

Innovative design approaches have further contributed to the advancement of power conversion systems. For example, researchers have developed compact and lightweight Stirling engines that are easier to integrate with PDCs. These designs also incorporate advanced cooling systems to prevent overheating and ensure stable operation under high-temperature conditions. Additionally, modular TEG designs have been developed, allowing for flexible configuration and scalability in solar power systems. Power conversion systems are a critical component of PDSCs, enabling the efficient conversion of concentrated solar energy into usable power. Technologies such as Stirling engines and TEGs have demonstrated significant potential for high-temperature applications, thanks to their high efficiency and adaptability. However, their integration with parabolic dish systems presents several challenges, including thermal losses, operational stability, and system compatibility. Recent innovations in hybrid systems, advanced materials, and novel design approaches have addressed many of these challenges, paving the way for more efficient and reliable power conversion technologies. As solar energy continues to play an increasingly important role in the global energy landscape, the development of advanced power conversion systems will be crucial for maximizing the potential of PDSCs. Future research should focus on further improving the efficiency, reliability, and scalability of these technologies to meet the growing demand for sustainable and high-performance solar power solutions.

5 Integrated System Redesign for Maximum Efficiency

PDSCs have emerged as a pivotal technology in harnessing solar energy for high-temperature applications. As the demand for more efficient and reliable solar energy systems grows, the need for integrated redesign approaches has become increasingly important. A holistic redesign incorporates multiple elements of the parabolic dish system, including advanced control algorithms, real-time monitoring, and optimized material selection, to ensure maximum efficiency and operational stability. This section delves into the theoretical and practical advancements in system integration, focusing on key aspects that collectively enhance the performance of parabolic dish systems. One crucial aspect of redesigning parabolic dish systems is the incorporation of advanced control algorithms. These algorithms enable precise tracking of the sun's position, ensuring optimal alignment between the dish and solar radiation throughout the day. The integration of predictive models that account for weather variations and solar intensity fluctuations can further improve tracking accuracy. Control algorithms also play a vital role in adjusting the receiver's orientation and optimizing heat transfer mechanisms, which are critical for maintaining high thermal efficiency. For example, the application of machine learning-based algorithms has shown promise in predicting solar irradiance patterns and adapting system operations accordingly [57].

Integrated redesign is essential for improving the efficiency, reliability, and long-term sustainability of PDSC systems as shown in Figure 8. A holistic redesign combines advanced monitoring and control technologies, optimized materials, improved optical configurations, enhanced thermal management, and robust structural designs. Real-time monitoring systems are particularly important because sensors and data-acquisition technologies can continuously measure critical operating parameters, including temperature, pressure, solar intensity, and optical alignment. Continuous feedback allows operators to detect deviations and performance problems rapidly and implement corrective actions. Sensors positioned on reflective surfaces can identify alignment changes resulting from wind loads or structural deformation, while Internet of Things (IoT) technologies enable remote monitoring, diagnostics, and control. These capabilities reduce dependence on on-site maintenance and improve operational reliability.

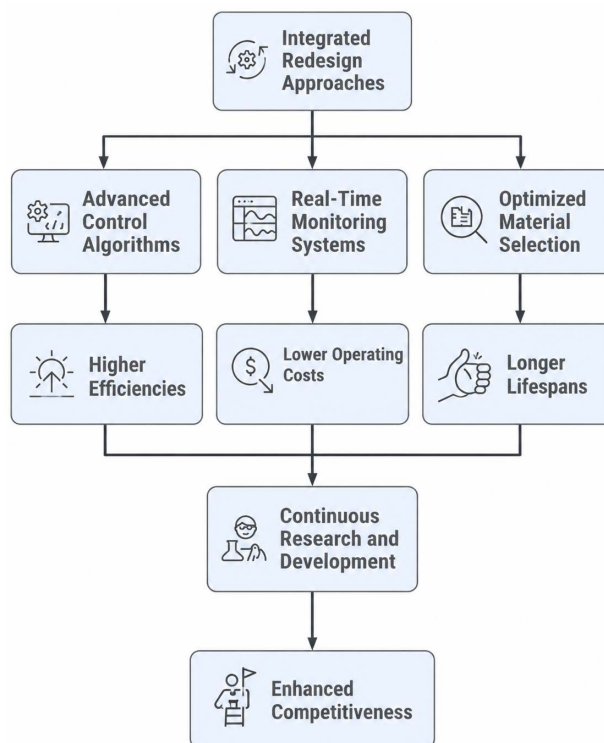


Figure 8. Integrated redesign for maximum efficiency for parabolic dish solar collators

Material selection is another important aspect of system redesign. Reflective materials with high optical reflectivity and durability can reduce radiation losses, while thermally resistant materials improve receiver performance under high-temperature operating conditions. Advanced coatings, including anti-soiling and self-cleaning surfaces, can reduce optical losses caused by dust accumulation, which is particularly important in arid and desert environments. Nanomaterials and advanced thermal coatings also offer opportunities to improve both optical and thermal performance. Energy losses in parabolic dish systems can generally be classified into optical, thermal, and mechanical losses. Optical losses occur when solar radiation is not accurately reflected and concentrated onto the receiver because

of surface imperfections, tracking errors, misalignment, or dust accumulation. Ray-tracing simulations provide an effective method for optimizing dish geometry, focal characteristics, and receiver positioning to maximize solar concentration. Automated cleaning systems and self-cleaning coatings can further maintain the optical quality of reflective surfaces and minimize losses associated with soiling. Thermal losses occur primarily during the transfer of concentrated solar energy to the working fluid and through radiation and convection from the receiver. Advanced cavity receivers can significantly reduce these losses by limiting direct exposure of the absorber to the surrounding environment. Improved heat-transfer materials and optimized receiver geometries can further enhance thermal performance. In addition, PCMs provide an effective TES solution because they can absorb and release substantial amounts of thermal energy at relatively constant temperatures. PCM integration can stabilize system operation and extend energy availability during periods of reduced solar irradiance. Mechanical losses are associated with structural deformation, wind-induced movement, vibration, and inaccurate tracking. Lightweight, high-strength materials can reduce structural loads and improve tracking performance, while advanced control algorithms can maintain accurate solar alignment under changing environmental conditions. Consequently, successful system integration requires simultaneous optimization of optical, thermal, and mechanical components rather than improving individual components independently.

Several recent developments demonstrate the potential of integrated redesign. Hybrid systems combining PDCs, high-temperature thermal storage, and Stirling engines have demonstrated the ability to maintain electricity generation beyond periods of direct solar availability. The integration of PCMs provides thermal buffering, while Stirling engines can efficiently convert high-temperature solar heat into electricity. Other applications have focused on industrial process heat, where redesigned receivers and advanced heat-transfer materials improve thermal delivery for high-temperature operations such as metal processing, smelting, and annealing. Real-time monitoring and control further improve reliability and reduce operational downtime. Parabolic dish systems have also been investigated for decentralized energy production in remote and off-grid communities. Lightweight and modular designs simplify transportation, installation, and maintenance, while IoT-enabled monitoring systems allow system performance to be supervised remotely. These features are particularly valuable in locations where conventional grid infrastructure and technical maintenance services are limited.

The integrated redesign represents a promising pathway for maximizing the performance of parabolic dish solar systems. Combining advanced optical optimization, real-time monitoring, intelligent control, improved receiver designs, TES, innovative materials, and structural optimization can substantially reduce energy losses while increasing efficiency, reliability, and operational flexibility. Continued research should focus on integrating these technologies into cost-effective and scalable systems capable of operating under harsh environmental conditions. Such developments can expand the application of parabolic dish technology from electricity generation to industrial process heat, energy storage, and decentralized energy supply, thereby strengthening its contribution to sustainable and reliable solar-energy systems.

6 Emerging Innovations and Technological Advancements

PDSCs are highly suitable for hybrid renewable-energy systems because they can concentrate solar radiation to achieve high temperatures and support efficient thermodynamic energy conversion. Their integration with complementary energy sources and storage technologies can overcome the intermittency of solar energy, improve energy reliability, and provide continuous power during periods of low solar irradiance or nighttime. Hybrid configurations therefore offer an important pathway toward more flexible, efficient, and sustainable energy systems.

One important configuration combines PDSCs with biomass energy. Solar energy can provide most of the required energy during periods of strong solar radiation, while biomass can supply energy when solar availability decreases. Demonstrated the potential of this approach through a large-scale system in Brack City, Libya, incorporating a 36,560 m³ biomass generator and a 6,006 m² PDSC collector field. The system generated an additional 5,670,534 kWh annually, illustrating the potential of solar–biomass integration to improve energy supply while reducing dependence on fossil fuels. PDSCs can also be integrated with wind energy because the two resources often have complementary availability. Solar energy is generally strongest during daylight hours, whereas wind resources can remain available during evening and nighttime periods. Numerical and experimental investigations, including the work of indicate that combining PDSCs with wind turbines can increase energy output and improve operational stability. Such systems reduce dependence on a single intermittent resource and can provide a more consistent renewable-energy supply.

Energy storage further strengthens the performance of PDSC-based hybrid systems. Batteries and TES can preserve excess energy and release it when solar radiation is insufficient. Demonstrated the value of thermal storage in parabolic dish–Stirling systems by maintaining power production during periods of reduced solar availability. Thermal storage can also reduce repeated thermal cycling, potentially improving system durability and operational life. The combination of PDSCs with storage is particularly important for applications requiring continuous and stable power.

Hybridization also provides significant benefits for industrial energy applications. The integration of solar

thermal energy into conventional steam power systems can improve overall energy efficiency and reduce greenhouse-gas emissions [11]. For example, solar heat supplied through PDSCs can be used for feedwater heating, reducing the amount of conventional fuel required. Similarly, hybrid PDSC systems can provide high-temperature heat for industrial processes while maintaining greater operational flexibility than standalone solar systems.

Another major advantage is the potential for increased energy production through multi-technology configurations. Other configurations combine PDSCs with PV cells and TEGs. In such systems, PV technology converts sunlight directly into electricity, PDSCs provide high-temperature thermal energy, and TEGs recover part of the available thermal energy for additional electrical production. This multi-conversion approach can increase the utilization of available solar energy. PDSC hybridization is also promising for remote and off-grid applications. Combining PDSCs with other solar concentration technologies, such as heliostat fields and central receivers, can provide reliable electricity to communities with limited access to conventional power grids. These systems can reduce dependence on fossil-fuel generators and contribute to improved energy access. Furthermore, PDSCs can support hydrogen production by supplying the high-temperature thermal energy required by thermochemical or hybrid hydrogen-production processes. The integration of thermal storage materials, including advanced nanoparticles such as Co_3O_4 , can further improve the stability and efficiency of these systems.

PDSC-based hybrid systems offer substantial advantages in reliability, efficiency, energy availability, environmental performance, and adaptability. Their integration with biomass, wind, PV, TEGs, thermal storage, steam systems, and hydrogen technologies enables more effective utilization of renewable resources and reduces the limitations associated with solar intermittency. Nevertheless, successful implementation requires careful optimization of system configuration, resource availability, thermal management, control strategies, storage capacity, and economic feasibility. Future research should therefore emphasize advanced control systems, high-performance thermal storage, intelligent energy management, techno-economic optimization, and integrated hybrid designs. Such developments can enhance the role of PDSCs in creating resilient, low-carbon energy systems for residential, industrial, and remote applications [58, 59].

7 Hydrogen Production and Energy Storage

PDCs offer significant potential for sustainable hydrogen production because of their ability to concentrate solar radiation and generate the high temperatures required for advanced thermochemical and high-temperature electrolysis (HTE) processes. By focusing sunlight onto a small receiver, PDC systems can achieve temperatures above 1,000 °C, providing substantial thermal energy that can reduce the electrical energy requirement of water electrolysis and improve the overall efficiency of hydrogen production as shown in Figure 9. This makes PDC technology particularly attractive for producing green hydrogen while reducing dependence on fossil-fuel-based energy. A major advantage of PDC-based hydrogen production is the possibility of integrating solar thermal energy with electricity and other useful outputs. Investigated a PDC-based multigeneration system capable of producing green hydrogen, electricity, and clean water. Such integrated systems demonstrate the flexibility of concentrated solar technology to address several energy and environmental challenges simultaneously. PDC systems are also highly scalable, ranging from small, decentralized units for remote communities to large collector fields for industrial hydrogen production [60, 61].



Figure 9. Hydrogen production and energy storage parabolic dish solar collectors (PDSCs)

However, the intermittent nature of solar radiation represents an important challenge. TES is therefore essential for maintaining stable operating temperatures and enabling hydrogen production beyond periods of peak solar availability. Molten salts are among the most promising storage media because of their high heat capacity and ability to operate at elevated temperatures. Stratified storage tanks can store excess heat during periods of strong solar

radiation and release it later, including during nighttime or cloudy conditions. This improves the continuity of HTE operation and increases the utilization of collected solar energy.

PCMs provide another attractive storage option. Unlike sensible-heat storage, PCMs store substantial amounts of energy as latent heat during phase transitions. Their high energy-storage density and relatively compact configuration make them suitable for PDC applications, particularly where space and system size are important considerations. PCM-based storage can provide thermal regulation and improve operational flexibility, supporting more stable hydrogen production under fluctuating solar conditions [62, 63]. In addition to thermal storage, efficient hydrogen storage is required for the successful deployment of solar-powered hydrogen systems. Technologies such as compressed hydrogen, metal hydrides, and chemical carriers can facilitate hydrogen storage, transportation, and integration with existing energy infrastructure. The combination of solar thermal generation, TES, and appropriate hydrogen-storage technologies can therefore create a more reliable and flexible renewable hydrogen supply chain. Several experimental and numerical studies have demonstrated the feasibility of PDC-based hydrogen systems. Investigated an integrated PDC and solar-pond configuration for hydrogen production and reported promising energetic and exergetic performance. Other studies have explored the combination of PDCs with linear Fresnel reflectors, demonstrating that hybrid solar-concentration technologies can improve thermal performance and reduce optical losses through complementary operation. These studies highlight the importance of system-level optimization rather than focusing solely on individual collector components.

Environmental performance is another important benefit. PDC-based hydrogen production can substantially reduce carbon emissions compared with conventional hydrogen production routes based on fossil fuels. The integration of renewable thermal energy and electricity can therefore contribute to the development of low-carbon hydrogen and support global decarbonization objectives. Thermal storage can further improve environmental performance by increasing the fraction of solar energy utilized and reducing the need for backup fossil-fuel generation. Despite these advantages, several technical and economic barriers remain. High initial capital costs for PDC collectors, receivers, tracking systems, and thermal storage can limit large-scale deployment. Receiver materials must also withstand extreme temperatures while maintaining adequate thermal conductivity, mechanical strength, and resistance to degradation. Advanced ceramics, coatings, and high-temperature composites are therefore important areas of research. Accurate solar tracking and alignment are equally critical because small optical errors can significantly reduce the concentration ratio and available thermal energy. Future developments should focus on modular and scalable PDC configurations, advanced TES materials, improved receiver designs, and intelligent control systems. Artificial intelligence and machine-learning techniques could optimize solar tracking, predict solar-resource availability, manage thermal storage, and dynamically control hydrogen production. Reducing manufacturing costs through advanced production methods and economies of scale will also be necessary for commercialization. PDC technology combined with HTE, TES, and advanced hydrogen-storage solutions represents a promising pathway for producing clean and reliable green hydrogen. Continued innovation, system integration, techno-economic optimization, and large-scale demonstration projects will be essential to overcome current limitations and establish PDC-based hydrogen systems as a practical component of future sustainable energy infrastructure [58, 59].

8 Decentralized Energy Supply for Remote Areas

The global energy demand continues to rise at an unprecedented rate, especially in remote and off-grid regions where access to conventional energy infrastructure remains challenging. PDSCs, as a CSP technology, have emerged as a promising solution to address this gap by providing decentralized energy supply. These systems leverage solar energy to generate electricity or thermal energy, offering a sustainable and environmentally friendly alternative to fossil fuels. This section explores the potential of PDSCs in delivering reliable energy solutions to remote areas, examines the design and operational challenges associated with decentralized energy systems, and highlights successful case studies showcasing their implementation.

PDSCs offer a promising solution to the persistent energy-access challenges faced by remote and off-grid communities. Geographical isolation, limited grid infrastructure, and dependence on expensive diesel generators often restrict reliable electricity access in these regions. PDSCs provide an alternative by directly converting abundant solar resources into useful thermal or electrical energy without requiring connection to centralized grids. Their compact configuration, high concentration ratio, modularity, and ability to achieve high operating temperatures make them particularly suitable for decentralized energy systems as shown in Table 4. One of the principal advantages of PDSCs is their ability to concentrate solar radiation onto a receiver, generating high-temperature heat that can be converted into electricity through technologies such as Stirling engines. The integration of PDSCs with Stirling engines and energy storage can provide reliable electricity for households, water pumping, agricultural activities, and small-scale industries. Hybrid renewable-energy systems can further improve reliability by combining solar energy with thermal storage or backup resources such as biomass. Since electricity is generated close to the point of consumption, decentralized PDSC systems can also reduce transmission losses and avoid the substantial costs associated with extending centralized electricity grids to isolated locations. The modularity and scalability of

PDSCs represent additional advantages for remote applications. Systems can be designed according to the energy requirements of individual households, small communities, or larger industrial facilities. This flexibility allows PDSCs to be adapted to different geographical conditions and energy demands. Their autonomous operation and potential for remote monitoring can further reduce maintenance requirements and improve long-term reliability.

Table 4. Decentralized energy supply for remote areas using parabolic dish solar collectors (PDSCs)

Aspect	Description	Key Benefits	Challenges
System configuration	Standalone or micro-grid PDSC systems coupled with Stirling engines and/or phase change material (PCM) storage	Independent power generation	Initial capital cost
Energy source	Direct Normal Irradiance (DNI)	Abundant in arid and remote regions	Weather dependency
Power output	Small-scale (1–25 kW) modular units	Scalable and flexible deployment	Limited nighttime generation without storage
Energy storage	PCM-based thermal storage or hybrid batteries	Dispatchable power, load shifting	Thermal management complexity
Reliability	Fewer moving parts (solar field side)	High system reliability	Maintenance of tracking system
Environmental impact	Zero emissions during operation	Reduced diesel dependency	Land and alignment requirements
Economic feasibility	Reduced fuel transport and costs	Long-term cost savings	Payback depends on DNI
Typical applications	Rural electrification, telecom towers, desalination, off-grid facilities	Energy security for remote communities	Skilled installation required

Despite these benefits, several technical, economic, and environmental challenges must be addressed. PDSC performance depends strongly on local solar irradiation, ambient temperature, wind conditions, and other climatic factors. Strong winds can cause structural deformation or damage to reflectors, requiring robust and lightweight designs. Accurate tracking and alignment are also essential because even small deviations can produce significant optical losses and reduce the amount of solar energy reaching the receiver. Therefore, reliable tracking systems and advanced control strategies are particularly important for maintaining performance under changing environmental conditions. Energy storage is another critical consideration. Because solar radiation is unavailable at night and may fluctuate during cloudy periods, TES can improve the continuity of energy supply. However, the development of efficient and affordable TES systems remains challenging in remote regions where advanced materials, technical expertise, and maintenance services may be limited. Economic barriers are also significant. High manufacturing and installation costs, together with the expense of precision tracking systems, receivers, and Stirling engines, can restrict adoption. These challenges can be addressed through low-cost manufacturing methods, locally available materials, government incentives, subsidies, technology-transfer programs, and partnerships among governments, research institutions, non-governmental organizations, and private companies.

Environmental conditions create further operational challenges. Dust accumulation on reflective surfaces can substantially reduce optical efficiency, particularly in arid and desert environments. Regular cleaning is therefore necessary, although maintenance can be difficult and expensive in isolated locations. Extreme weather conditions, including heavy rainfall, strong winds, or snow, may also affect structural integrity and system performance. Consequently, PDSCs for remote applications should incorporate durable materials, protective designs, automated cleaning, and remote monitoring technologies.

Several applications demonstrate the potential of PDSCs for decentralized energy supply. Solar-powered microgrids integrating PDSCs and Stirling engines can provide electricity for lighting, water pumping, and local industries while reducing dependence on diesel generators. Thermal storage can ensure continued operation during nighttime and periods of reduced solar radiation. PDSCs can also support water desalination in remote coastal areas by supplying the thermal energy required for processes such as humidification–dehumidification desalination. This combination of renewable energy and water production is particularly valuable in regions facing both energy and water shortages. PDSC-based solar cooking systems with thermal storage can provide clean cooking energy for remote communities. By reducing dependence on traditional biomass fuels, these systems can contribute to lower deforestation, reduced indoor air pollution, and improved public health. Overall, PDSCs have considerable potential to enhance energy access and support sustainable development in remote and off-grid regions. Continued progress in low-cost manufacturing, durable materials, intelligent tracking, thermal storage, and remote monitoring will be

essential for overcoming current barriers and enabling wider deployment. Through appropriate system design and supportive policies, PDSCs can provide reliable, clean, and locally generated energy while reducing dependence on fossil fuels and strengthening the resilience of underserved communities.

9 Numerical Modeling and Experimental Investigations

The performance of PDSCs has garnered significant attention due to their potential in high-temperature applications. To optimize their design and performance, researchers rely heavily on simulation techniques that predict system behavior under various operating conditions. This section delves into the use of CFD and ray-tracing techniques as key tools for modeling PDSCs. It also analyzes the accuracy and limitations of existing simulation tools and highlights studies that demonstrate how these techniques have led to significant design improvements as shown in Table 5.

Table 5. Numerical modeling and experimental investigations on the performance of parabolic dish solar collectors (PDSCs) for high-temperature applications

Aspect	Numerical Modeling Studies	Experimental Investigations
Primary objective	Predict thermal, optical, and electrical performance; optimize design parameters	Validate models; assess real-world performance and losses
Modeling/setup	Computational fluid dynamics (CFD), ray-tracing, Monte Carlo methods	Prototype PDSCs, outdoor/indoor test rigs
Parameters studied	Concentration ratio, receiver geometry, heat transfer coefficients, material properties	Solar irradiance, receiver temperature, thermal efficiency
Receiver types	Cavity, tubular, volumetric, phase change material (PCM)-integrated receivers	Cavity and tubular receivers most common
Thermal performance	High predicted efficiencies (70–90%) under ideal conditions	Slightly lower efficiencies due to optical and thermal losses
Optical loss analysis	Detailed evaluation of reflectivity, interception, and spillage losses	Measured optical efficiency and tracking errors
Heat transfer analysis	Conduction, convection, radiation modeled in detail	Temperature distribution and heat losses measured
Energy output	Thermal output and power generation potential estimated	Actual thermal/electrical output recorded
Advantages	Cost-effective optimization; parametric studies	Realistic validation; reliability assessment
Limitations	Assumptions and boundary condition sensitivity	High cost, weather dependency, limited scalability
Overall contribution	Design optimization and performance prediction	Model validation and system feasibility

10 Conclusion

PDSCs represent a highly promising CSP technology for high-temperature energy applications, offering superior optical efficiency, high concentration ratios, and flexibility in power conversion integration. This comprehensive review has examined the fundamental design principles of PDSCs, including concentrator geometry, optical performance, receiver configurations, and solar tracking systems, highlighting their critical role in achieving efficient solar-to-thermal energy conversion. The analysis demonstrates that precise concentrator design, advanced reflective materials, and accurate dual-axis tracking are essential for minimizing optical losses and maximizing system efficiency. Receiver technology remains a central factor in determining thermal performance. Cavity receivers, tubular receivers, volumetric absorbers, and PCM-integrated designs each present distinct advantages and limitations depending on operating temperature, heat transfer requirements, and application context. The integration of PCMs and advanced cooling strategies has shown strong potential in enhancing thermal stability, reducing losses, and improving dispatchability. Furthermore, the coupling of PDSCs with high-efficiency power conversion systems particularly Stirling engines and hybrid Stirling–thermoelectric configurations—enables effective utilization of high-temperature solar energy for electricity generation and industrial process heat. System-level optimization, including real-time monitoring, advanced control algorithms, and integrated redesign approaches, has emerged as a key pathway

toward improving operational reliability and long-term performance. Recent advancements in hybrid systems, TES, and multi-energy integration further expand the applicability of PDSCs for decentralized power generation, industrial heating, and solar-driven hydrogen production.

Despite their significant potential, challenges related to material durability, thermal losses, system cost, and large-scale deployment remain. Addressing these challenges will require continued advancements in materials science, receiver design, thermal management, and intelligent control systems, supported by robust numerical modeling and experimental validation. Overall, PDSCs are well positioned to play a vital role in future high-temperature renewable energy systems, contributing to sustainable energy transitions and the decarbonization of power generation and industrial sectors.

Author Contributions

Conceptualization, S.A.H.; methodology, S.A.H.; formal analysis, S.A.H.; writing—original draft preparation, S.A.H.; supervision, M.J.C.; resources, M.J.C.; project administration, M.J.C. and R.A.S.; writing—review and editing, R.A.S.; validation, R.A.S. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon reasonable request.

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Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] T. Mancini, P. Heller, B. Butler, B. Osborn, W. Schiel, V. Goldberg, R. Buck, R. Diver, C. Andraka, and J. Moreno, “Dish-stirling systems: An overview of development and status,” *J. Sol. Energy Eng.*, vol. 125, no. 2, pp. 135–151, 2003. <https://doi.org/10.1115/1.1562634>
- [2] J. A. Alarcón, J. E. Hortúa, and A. Lopez G., “Design and construction of a solar collector parabolic dish for rural zones in Colombia,” *Tecciencia*, vol. 7, no. 14, pp. 14–22, 2013. <https://doi.org/10.18180/tecciencia.2013.14.2>
- [3] S. S. A. Al-Hashmi, “Design, fabrication and testing of hybrid parabolic dish concentrator, Stirling engine & PCM-storage in Oman,” Ph.D. dissertation, University of Leeds, U.K., 2015. <https://etheses.whiterose.ac.uk/id/eprint/10417/>
- [4] S. Al-Hashmi, M. Chen, and S. Al-Saidi, “Advancing sustainable energy solutions for hot regions: An in-depth exploration of solar thermal energy storage (STES) technologies and applications,” *Eng. Res. Express*, vol. 7, no. 1, p. 012101, 2025. <https://doi.org/10.1088/2631-8695/adb8a0>
- [5] S. Al-Hashmi, M. Chen, N. Ahmad, and A. Tantray, “Hybrid solar power system using stirling engine and phase change material for energy storage: Design, optimization and efficiency enhancement,” *Front. Energy Res.*, vol. 14, p. 1720534, 2026. <https://doi.org/10.3389/fenrg.2026.1720534>
- [6] J. García-Ferrero, M. J. Santos, A. Medina, and A. C. Hernández, “Promising research trends for solar parabolic dish collectors,” *Appl. Therm. Eng.*, vol. 278, p. 126934, 2025. <https://doi.org/10.1016/j.applthermaleng.2025.126934>
- [7] J. García-Ferrero, R. P. M. Corral, J. A. Moctezuma-Hernández, D. Pérez-Gallego, S. Anvari, J. González-Ayala, A. Calvo-Hernández, J. M. M. Roco, M. J. Santos, and A. Medina, “Parabolic dish collectors for concentrated solar power: A comprehensive review on their subsystems and overall integration,” *Energies*, vol. 18, no. 24, p. 6596, 2025. <https://doi.org/10.3390/en18246596>
- [8] Y. Zhi, S. S. Liu, B. Yang, and X. H. Yu, “Analysis of the thermal performance of parabolic solar dish collector using MCRT-FVM coupled optical-thermal model,” *Energy*, vol. 322, p. 135579, 2025. <https://doi.org/10.1016/j.energy.2025.135579>
- [9] G. Barreto and P. Canhoto, “Modelling of a Stirling engine with parabolic dish for thermal to electric conversion of solar energy,” *Energy Convers. Manag.*, vol. 132, pp. 119–135, 2017. <https://doi.org/10.1016/j.enconman.2016.11.011>

- [10] A. Parveh and M. Baneshi, "Transient modelling and economic viability of parabolic dish concentrating photovoltaic-thermal (PD-CPVT) systems across diverse climatic zones," *Case Stud. Therm. Eng.*, vol. 74, p. 106949, 2025. <https://doi.org/10.1016/j.csite.2025.106949>
- [11] N. Nasiri, M. Mohammadzahari, R. M. Rafikh, M. Farrizi, H. Nasiri, I. Al-Busaidi, and S. Al-Hashmi, "Harnessing wind energy for a green hydrogen revolution in Oman: Potential, challenges, and economic viability," *J. Eng. Res.*, vol. 22, no. 1, pp. 55–67, 2025. <https://doi.org/10.53540/1726-6742.1309>
- [12] T. U. Shinde, V. H. Dalvi, R. G. Patil, C. S. Mathpati, S. V. Panse, and J. B. Joshi, "Thermal performance analysis of novel receiver for parabolic trough solar collector," *Energy*, vol. 254, p. 124343, 2022. <https://doi.org/10.1016/j.energy.2022.124343>
- [13] S. Dixit, "Solar technologies and their implementations: A review," *Mater. Today Proc.*, vol. 28, pp. 2137–2148, 2020. <https://doi.org/10.1016/j.matpr.2020.04.134>
- [14] F. Kibrete, T. T. Feisa, D. E. Woldemichael, and B. G. Mekonnen, "Vibration-based condition monitoring and fault diagnosis of rotating machinery: Fundamentals, signal processing techniques, standards, and emerging trends," *Results Eng.*, vol. 32, p. 112372, 2026. <https://doi.org/10.1016/j.reneng.2026.112372>
- [15] B. Kongtragool and S. Wongwiset, "A review of solar-powered Stirling engines and low temperature differential Stirling engines," *Renew. Sustain. Energy Rev.*, vol. 7, no. 2, pp. 131–154, 2003. [https://doi.org/10.1016/S1364-0321\(02\)00053-9](https://doi.org/10.1016/S1364-0321(02)00053-9)
- [16] E. Gholamalizadeh and J. D. Chung, "Exergy analysis of a pilot parabolic solar dish-stirling system," *Entropy*, vol. 19, no. 10, p. 509, 2017. <https://doi.org/10.3390/e19100509>
- [17] H. Chayet, I. Lozovsky, O. Kost, R. Loeckenhoff, and K. D. Rasch, "High efficiency, low cost parabolic dish system for cogeneration of electricity and heat," in *AIP Conference Proceedings*, 2010, pp. 175–178. <https://doi.org/10.1063/1.3509183>
- [18] S. S. Al-Hashmi, M. Chen, and S. S. Al-Saidi, "Optimizing thermal conductivity in latent heat energy storage systems using perforated fins: Experimental investigation on a vertical shell-and-tube heat exchanger," in *Recent Research on Environmental Earth Sciences, Geomorphology, Soil Science, Paleoclimate, and Karst (MedGU 2023)*, 2026, p. 489–494. https://doi.org/10.1007/978-3-032-20990-0_95
- [19] S. P. Badgujar, C. S. Kumar, and H. K. Wagh, "Optimizing solar PV panel performance through phase change material cooling: An experimental study," *Int. J. Mech. Eng.*, vol. 11, no. 8, pp. 32–38, 2024. <https://doi.org/10.14445/23488360/ijme-v11i8p104>
- [20] Q. Alkhalaf, A. R. S. Suri, S. S. Chandel, S. Thapa, and M. S. Al Ansari, "Performance investigation of a Scheffler solar cooking system combined with Stirling engine," *Mater. Today Proc.*, 2023. <https://doi.org/10.1016/j.matpr.2023.01.399>
- [21] Q. Alkhalaf, A. S. Bisht, A. R. S. Suri, and S. S. Chandel, "Modelling and simulation of concentrated solar powered Stirling engine based water pumping system," in *Challenges and Opportunities in Industrial and Mechanical Engineering: A Progressive Research Outlook*. CRC Press, 2024, pp. 987–993. <https://doi.org/10.1201/9781032713229-111>
- [22] M. Rahbari, P. Esfanjani, A. Mahmoudi, and M. S. Valipour, "Performance enhancement of a parabolic dish collector with a novel receiver assisted with a secondary reflector," *Sol. Energy Mater. Sol. Cells*, vol. 283, p. 113464, 2025. <https://doi.org/10.1016/j.solmat.2025.113464>
- [23] W. Ciciora, J. Farmer, D. Large, and M. Adams, "Signal reception," in *Modern Cable Television Technology (2nd ed.)*. Morgan Kaufmann, 2004, pp. 287–332. <https://doi.org/10.1016/B978-155860828-3/50009-6>
- [24] E. Bellos, "Progress in beam-down solar concentrating systems," *Prog. Energy Combust. Sci.*, vol. 97, p. 101085, 2023. <https://doi.org/10.1016/j.pecs.2023.101085>
- [25] S. Al-Hashmi, M. Chen, N. Ahmad, M. Abedi, R. Al Saadi, A. Al-Rajhi, S. Rekik, J. H. R. Mensah, and M. Khaleel, "Toward a sustainable energy future in the Gulf region: Overcoming barriers and enhancing renewable energy strategies," *Energy Sources, Part B*, vol. 21, no. 1, p. 2703866, 2026. <https://doi.org/10.1080/15567249.2026.2703866>
- [26] R. M. Elavarasan, V. Mudgal, L. Selvamanoahar, K. Wang, G. Huang, G. Shafiullah, C. N. Markides, K. Reddy, and M. Nadarajah, "Pathways toward high-efficiency solar photovoltaic thermal management for electrical, thermal and combined generation applications: A critical review," *Energy Convers. Manag.*, vol. 255, p. 115278, 2022. <https://doi.org/10.1016/j.enconman.2022.115278>
- [27] B. Nagappan, K. Mehar, S. P. Jena, A. Tiwari, K. V. R., A. Kushwah, K. P. Kumar, and A. Samrat, "Hybrid solar-phase change material energy-storage systems for low-carbon energy infrastructure: Design strategies, performance insights, and sustainability pathways," *Energy Convers. Manag.: X*, vol. 29, p. 101466, 2026. <https://doi.org/10.1016/j.ecmx.2025.101466>
- [28] S. D. Nazemi and M. Boroushaki, "Design, analysis and optimization of a solar dish/stirling system," *Int. J. Renew. Energy Dev.*, vol. 5, no. 1, pp. 33–42, 2016. <https://doi.org/10.14710/IJRED.5.1.33-42>

- [29] H. H. Al-Kayiem (Ed.), *The Future of Solar Sower*. Nova Science Publishers, 2023. <https://doi.org/10.52305/GNEE3683>
- [30] M. I. Alam, M. M. Nuhash, A. Zihad, T. H. Nakib, and M. M. Ehsan, "Conventional and emerging CSP technologies and design modifications: Research status and recent advancements," *Int. J. Thermofluids*, vol. 20, p. 100406, 2023. <https://doi.org/10.1016/j.ijft.2023.100406>
- [31] B. Kalidasan, M. A. Hassan, A. K. Pandey, and S. Chinnasamy, "Linear cavity solar receivers: A review," *Appl. Therm. Eng.*, vol. 221, p. 119815, 2023. <https://doi.org/10.1016/j.applthermaleng.2022.119815>
- [32] S. Al-Hashmi and M. Chen, "Thermal analysis of phase change materials storage in solar concentrator," *J. Energy Syst.*, vol. 7, no. 3, pp. 302–314, 2023. <https://doi.org/10.30521/jes.1082104>
- [33] Y. L. He, Y. Qiu, K. Wang, F. Yuan, W. Q. Wang, M. J. Li, and J. Q. Guo, "Perspective of concentrating solar power," *Energy*, vol. 198, p. 117373, 2020. <https://doi.org/10.1016/j.energy.2020.117373>
- [34] Y. L. He, S. Du, and S. Shen, "Advances in porous volumetric solar receivers and enhancement of volumetric absorption," *Energy Rev.*, vol. 2, no. 3, p. 100035, 2023. <https://doi.org/10.1016/j.enrev.2023.100035>
- [35] E. Bellos, "Progress in the design and the applications of linear fresnel reflectors—A critical review," *Therm. Sci. Eng. Prog.*, vol. 10, pp. 112–137, 2019. <https://doi.org/10.1016/j.tsep.2019.01.014>
- [36] M. I. Khan, F. Asfand, and S. G. Al-Ghamdi, "Progress in research and technological advancements of commercial concentrated solar thermal power plants," *Sol. Energy*, vol. 249, pp. 183–226, 2023. <https://doi.org/10.1016/j.solener.2022.10.041>
- [37] A. Madhlopa, "Classification of solar receivers," in *Solar Receivers for Thermal Power Generation*. Academic Press, 2022, pp. 61–78. <https://doi.org/10.1016/B978-0-323-85271-5.00001-X>
- [38] R. Riedel (Ed.), *Handbook of Ceramic Hard Materials*. Wiley-VCH, 2008. <https://doi.org/10.1002/9783527618217>
- [39] R. A. Shishkin, "Oxide-bonded silicon carbide and alumina ceramics obtained from template SCS powders," *Ceram. Int.*, vol. 51, no. 8, pp. 10340–10350, 2025. <https://doi.org/10.1016/j.ceramint.2024.12.467>
- [40] H. Togun, A. Basem, M. J. Jweeg, H. I. Mohammed, A. M. Abed, A. E. Anqi, A. Chattopadhyay, and N. Biswas, "Development and innovation using PCM in PV cooling systems: Passive and active approaches," *J. Therm. Anal. Calorim.*, vol. 150, no. 14, pp. 10725–10760, 2025. <https://doi.org/10.1007/s10973-025-14388-1>
- [41] A. Hassabou, A. Abotaleb, A. Abdallah, T. Klemm, and O. Andersen, "Passive cooling of photovoltaic modules in Qatar by utilizing PCM-matrix absorbers," in *Proceedings of the ISES Solar World Congress 2019 and IEA SHC International Conference on Solar Heating and Cooling for Buildings and Industry 2019*, Santiago, Chile, 2020, pp. 795–803. <https://doi.org/10.18086/swc.2019.16.03>
- [42] A. Durez, M. Ali, A. Waqas, K. Nazir, and S. Kumarasamy, "Modelling and optimization of phase change materials (PCM)-based passive cooling of solar PV panels in multi climate conditions," *Front. Energy Res.*, vol. 11, p. 1121138, 2023. <https://doi.org/10.3389/fenrg.2023.1121138>
- [43] A. Nandi and N. Biswas, "Enhanced melting dynamics of phase change material (PCM) based energy storage system combining modified fin and nanoparticles under solar irradiation," *Int. J. Numer. Methods Heat Fluid Flow*, vol. 35, no. 6, pp. 2104–2143, 2025. <https://doi.org/10.1108/hff-08-2024-0643>
- [44] Y. Zhang, C. Liao, J. Qin, Y. Bai, C. Liu, K. Zhen, F. Bai, and J. Wang, "Selection of high-performance thermal interface materials for space payload applications," *Case Stud. Therm. Eng.*, vol. 71, p. 106203, 2025. <https://doi.org/10.1016/j.csite.2025.106203>
- [45] K. Wang, D. Wang, B. Zhou, X. Wang, and T. Chen, "Study on the synergistic extinguishing effect of perfluorohexanone-nitrogen," *Case Stud. Therm. Eng.*, vol. 71, p. 106215, 2025. <https://doi.org/10.1016/j.csite.2025.106215>
- [46] S. F. Li, Z. H. Liu, and X. J. Wang, "A comprehensive review on positive cold energy storage technologies and applications in air conditioning with phase change materials," *Appl. Energy*, vol. 255, p. 113667, 2019. <https://doi.org/10.1016/j.apenergy.2019.113667>
- [47] E. Oró, A. De Gracia, A. Castell, M. M. Farid, and L. F. Cabeza, "Review on phase change materials (PCMs) for cold thermal energy storage applications," *Appl. Energy*, vol. 99, pp. 513–533, 2012. <https://doi.org/10.1016/j.apenergy.2012.05.022>
- [48] S. K. Vishnu and R. Senthil, "Review of key factors for optimizing the thermal performance of parabolic dish solar collectors," *Renew. Sustain. Energy Rev.*, vol. 226, p. 116348, 2026. <https://doi.org/10.1016/j.rser.2025.116348>
- [49] T. Al-Abri, M. Chen, M. R. Nikoo, S. Al-Hashmi, and A. Al-Hinai, "Economic analysis of blue and green hydrogen production in Oman: Comparison of various energy sources mix," *Energy Ecol. Environ.*, vol. 10, no. 2, pp. 225–242, 2025. <https://doi.org/10.1007/s40974024003419>
- [50] S. Kumar and A. K. Roy, "Advanced thermal management technologies for high-power electronic devices:

A review of passive, active, and hybrid cooling strategies, materials innovations, and AI-driven design for data-center, EV, and aerospace electronics,” *Next Mater.*, vol. 13, p. 103341, 2026. <https://doi.org/10.1016/j.nxmater.2026.103341>

- [51] R. Terdalkar, D. Doupis, M. Clark, A. Joshi, and C. Wang, “Transient simulation of high temperature high pressure solar tower receiver,” *Energy Procedia*, vol. 69, pp. 1451–1460, 2015. <https://doi.org/10.1016/j.egypro.2015.03.093>
- [52] S. Shahrzad and B. Umberto, “Parametric optimization of multifunctional integrated climate-responsive opaque and ventilated façades using CFD simulations,” *Appl. Therm. Eng.*, vol. 204, p. 117923, 2022. <https://doi.org/10.1016/j.applthermaleng.2021.117923>
- [53] J. A. Agwupuye, R. N. Ekere, J. N. Ihedioha, and M. C. Agwupuye, “Smart composite systems for integrated environmental sensing and remediation: Recent advances and future frontiers,” *Next Mater.*, vol. 13, p. 103337, 2026. <https://doi.org/10.1016/j.nxmater.2026.103337>
- [54] R. K. Mohapatra and Y. Choi, “Exploring multi-omics approaches in anammox-based wastewater treatment processes: A review of recent applications and technological advances,” *J. Environ. Manage.*, vol. 404, p. 129342, 2026. <https://doi.org/10.1016/j.jenvman.2026.129342>
- [55] A. Batooei, A. Qasemian, and A. Amiri, “Thermal analysis and efficiency optimization of diesel-stirling combined cycle with CI engine exhaust heat recovery,” *J. Environ. Manage.*, vol. 404, p. 129175, 2026. <https://doi.org/10.1016/j.jenvman.2026.129175>
- [56] M. Sreenivasulu, G. A. Khouqeer, N. AbdelAll, A. N. Alodhayb, N. P. Shetti, and T. M. Aminabhavi, “Environmentally benign visible-light-driven Fe-WSe₂ photocatalyst for hydrogen production and dye degradation,” *J. Environ. Manage.*, vol. 404, p. 129346, 2026. <https://doi.org/10.1016/j.jenvman.2026.129346>
- [57] Z. Ghahari, S. Baradaran, and H. Fatoorehchi, “Energy-efficiency with enhanced process safety in propylene-propane separation via multi-objective optimization and redesign,” *Energy Rep.*, vol. 15, p. 109020, 2026. <https://doi.org/10.1016/j.egy.2025.109020>
- [58] X. Wang, J. Shang, Z. Luo, L. Tang, X. Zhang, and J. Li, “Reviews of power systems and environmental energy conversion for unmanned underwater vehicles,” *Renew. Sustain. Energy Rev.*, vol. 16, no. 4, pp. 1958–1970, 2012. <https://doi.org/10.1016/j.rser.2011.12.016>
- [59] M. S. Hossain, A. K. Pandey, J. Tunio, N. Selvaraj, and N. A. Rahim, “The hybrid solar power/wind system for energy production, observation, application, and simulation,” in *Clean Energy for Sustainable Development: Comparisons and Contrasts of New Approaches*. Academic Press, 2017, pp. 337–368. <https://doi.org/10.1016/B9780128054239.000107>
- [60] L. T. C. Montoya, S. Lain, M. Issa, and A. Ilinca, “Renewable energy systems,” in *Hybrid Renewable Energy Systems and Microgrids*. Academic Press, 2021, pp. 103–177. <https://doi.org/10.1016/B978-0-12-821724-5.00013-1>
- [61] Y. S. Mohammed, M. W. Mustafa, and N. Bashir, “Hybrid renewable energy systems for off-grid electric power: Review of substantial issues,” *Renew. Sustain. Energy Rev.*, vol. 35, pp. 527–539, 2014. <https://doi.org/10.1016/j.rser.2014.04.022>
- [62] D. Surroop, “Energy transition to decarbonize the energy system in Mauritius,” *Curr. Opin. Environ. Sustain.*, vol. 61, p. 101268, 2023. <https://doi.org/10.1016/j.cosust.2023.101268>
- [63] Y. Yuan, J. Wang, X. Yan, B. Shen, and T. Long, “A review of multi-energy hybrid power system for ships,” *Renew. Sustain. Energy Rev.*, vol. 132, p. 110081, 2020. <https://doi.org/10.1016/j.rser.2020.110081>